

Two polynomial-size dice in a completely finite fair family

Complete proof; independently reviewed

30 September 2026

Scope. This proof combines the reviewed real step-rank construction and positive completion with a rational smoothing argument and the companion clustered lifting lemma. All parts have passed full independent mathematical reviews by the root, rational-design, and independent-direction agents. This is not formal verification. The other dice may be extremely large.

Theorem 1. *There are absolute constants C, D such that for every $n \geq 3$ a completely finite permutation-fair family of n equiprobable dice has two specified dice with at most Cn^D faces each. The other $n - 2$ dice can have a common finite face count. The proof below permits $D = 162$; no attempt is made to optimize it.*

Put $m = n - 2$. The two reviewed inputs are `../rational_designs/asymptotic/two_discrete_step_compatibility.tex` and `../independent_directions/two_exceptional_positive_completion.tex`. They supply distinct nodes $0 < b_1 < \dots < b_K < 1$ and increasing rational ranks $0 < r_1 < \dots < r_K < 1$, with

$$\frac{1}{K} \sum b_q^j = \frac{1}{j+1} \quad (0 \leq j \leq m+1), \quad \frac{1}{K} \sum r_q b_q^j = \frac{1}{j+2} \quad (0 \leq j \leq m), \quad (1)$$

where $K = O((m+1)^4)$ and the r_q have a common denominator $L = O((m+1)^{160})$. Their closeness guarantees a positive completion f_0 , with $1/2 \leq f_0 \leq 3/2$, on the point-node gaps.

We also use a property proved in the step-rank construction: with r fixed, the Jacobian in b of the $m+1$ ordinary moment equations and the m nonconstant rank-weighted equations has row rank $2m+1$. Indeed, before the small Newton correction its rows evaluate $\mathcal{P}_m + (r-b)\mathcal{P}_{m-1}$, whose Gram matrix is positive by the Legendre conditioning lemma. The fixed-right-inverse Newton map in that proof is a contraction on its affine correction subspace. At its solution, therefore, the right inverse composed with the new Jacobian is invertible on that subspace. This directly preserves row rank $2m+1$; no further quantitative lower bound is needed.

Lemma 1 (Rational positive smoothing of the discrete variable). *There are disjoint rational closed intervals $J_q \subset (0, 1)$ and a positive affine density on each J_q , zero elsewhere, defining a probability measure β , such that*

$$\beta(J_q) = 1/K, \quad (2)$$

$$\int x^j d\beta(x) = 1/(j+1) \quad (0 \leq j \leq m+1), \quad (3)$$

$$\sum_q r_q \int_{J_q} x^j d\beta(x) = 1/(j+2) \quad (0 \leq j \leq m). \quad (4)$$

The conditional density in every box, rescaled to $[-1, 1]$, is between $1/4$ and $3/4$. All density coefficients are rational. The boxes may be made arbitrarily close to the points b_q and arbitrarily short.

Proof. Let $\varepsilon \downarrow 0$ through positive rational numbers and choose rational centers c_q with $|c_q - b_q| \leq \varepsilon^3$. Set $J_q = [c_q - \varepsilon, c_q + \varepsilon]$; these are disjoint and interior for small enough ε . On J_q use density

$$\frac{1}{2K\varepsilon} \left(1 + a_q \frac{x - c_q}{\varepsilon} \right).$$

Its box mass is $1/K$ for all a_q . At $a = 0$, the residual in the $2m + 1$ nonconstant constraints (3)–(4) is $O_m(\varepsilon^2)$ by symmetric Taylor expansion and (1). The coefficient matrix for the variables a_q , divided by ε , converges to one third of the full-rank Jacobian specified above. This follows from

$$\frac{1}{2} \int_{-1}^1 y P(c_q + \varepsilon y) dy = \frac{\varepsilon}{3} P'(b_q) + O_m(\varepsilon^3).$$

Choose a nonsingular square minor of that Jacobian, set the other a_q to zero, and solve the corresponding linear system exactly. It gives $a_q = O_m(\varepsilon)$, hence $|a_q| \leq 1/2$ for sufficiently small boxes. All entries of the actual system and its targets are rational, so all a_q are rational. The two degree-zero constraints are already automatic from the box masses and $K^{-1} \sum r_q = 1/2$. This proves the lemma. $\square$

Lemma 2 (Positive rational completion in the complementary gaps). *For all sufficiently short boxes in the preceding lemma, there is a rational piecewise polynomial density f , supported on their complement, with mass $r_{q+1} - r_q$ on the q th complementary gap, where $r_0 = 0, r_{K+1} = 1$. On every such gap, $1/4 \leq f \leq 7/4$. If A has density f , B has law β , and m further variables are independent uniforms, all complete orders are uniform.*

Proof. Write $G_\varepsilon = [0, 1] \setminus \bigcup_q J_q$ and define

$$S_j^\varepsilon(a) = \int_{b > a} b^j d\beta(b), \quad 0 \leq j \leq m.$$

On each component of G_ε these are constant. Let

$$V_\varepsilon = \text{span}\{a^j : 0 \leq j \leq m\} + \text{span}\{a^r S_s^\varepsilon(a) : r + s \leq m\}, \quad S_\varepsilon = \text{span}\{1, S_0^\varepsilon, \dots, S_m^\varepsilon\},$$

where the spaces are restricted to G_ε . Let P_ε be averaging separately on its complementary gaps.

The matrix

$$M_{q,s}^\varepsilon = \int_{J_q} b^s d\beta(b), \quad 0 \leq s \leq m,$$

has full column rank for sufficiently small boxes, since it converges to the scaled Vandermonde matrix $K^{-1}(b_q^s)$. Consequently

$$V_\varepsilon \cap \text{ran } P_\varepsilon = S_\varepsilon. \quad (5)$$

To see this directly, write a function in V_ε as $p(a) + \sum_{r+s \leq m} c_{r,s} a^r S_s^\varepsilon(a)$. If it is constant in each gap, the coefficient of a^r , for every $r \geq 1$, vanishes in all gaps. Subtracting consecutive gaps yields $\sum_s c_{r,s} M_{q,s}^\varepsilon = 0$ for all q . The full-rank property, also for each initial set of columns, gives $c_{r,s} = 0$ and then the coefficient of a^r in p is zero. Only the displayed step space remains. The same argument, including $r = 0$, proves independence of the indicated polynomial/step basis.

Let g_ε be constant on each gap with the prescribed mass $\gamma_q = r_{q+1} - r_q$. Seek a zero-gap-mean polynomial correction h_ε in $W_\varepsilon = (I - P_\varepsilon)V_\varepsilon$ such that

$$\int_{G_\varepsilon} h_\varepsilon F = \int_0^1 F(a) da - \int_{G_\varepsilon} g_\varepsilon F \quad (F \in V_\varepsilon). \quad (6)$$

For the full-interval integral, use the functions before restriction. The right side annihilates S_ε : the constant is immediate, and Fubini, the gap masses, (3), and (4) give

$$\int_{G_\varepsilon} g_\varepsilon S_j^\varepsilon = \sum_q r_q M_{q,j}^\varepsilon = \frac{1}{j+2} = \int_0^1 S_j^\varepsilon(a) da.$$

Thus (5) makes (6) a well-defined Riesz problem on W_ε .

Here is the needed continuity without any uniform lower bound on box widths. A basis of W_ε is given by applying $I - P_\varepsilon$ to a^r ($1 \leq r \leq m$) and to $a^r S_s^\varepsilon$ ($r \geq 1$, $r + s \leq m$). The preceding full-rank argument proves its independence. Its Gram matrix and the right-hand vector in (6) converge as $\varepsilon \rightarrow 0$ to those of the point-node problem in the reviewed positive-completion proof. That limiting Gram matrix is positive definite. Therefore the coefficients of h_ε converge to those of its point-node representative. After identifying each moving gap affinely with its original gap, all functions involved are polynomials of bounded degree with convergent coefficients. Convergence is uniform on each closed gap. The limiting completed density is the reviewed $f_0 \in [1/2, 3/2]$. Hence $f = g_\varepsilon + h_\varepsilon \in [1/4, 7/4]$ on all gaps for sufficiently small boxes. All Gram entries and right-hand entries are rational: box and gap endpoints, ranks, and polynomial density coefficients are rational. Thus the coefficients of f are rational as well.

By (6), f integrates every test in V_ε exactly as Lebesgue measure does. For $r + s \leq m$,

$$\mathbb{E}[A^r B^s \mathbf{1}_{A < B}] = \int f(a) a^r S_s^\varepsilon(a) da = \frac{1}{r+1} \int b^{r+s+1} d\beta(b) = \frac{1}{(r+1)(r+s+2)}.$$

The ordinary moments and the opposite triangle follow in the same way as in the reviewed point-node completion. These are all the polynomial chamber moments needed for m other uniform variables, proving full order fairness. $\square$

Proof of Theorem 1. We have rational polynomial densities for A and B on alternating, disjoint interval interiors. Their cluster masses have denominators L and K , respectively. The conditional densities, after rescaling a cluster to $[0, 1]$, are bounded above and below by absolute positive constants: for A use the bounds $1/4, 7/4$ and its cell average; for B use its affine density bounds.

Choose an integer $T = \lceil C(m+1)^2 \rceil$ with sufficiently large absolute C . On each B box take an equal-weight quadrature with exactly T nodes. On an A gap of mass a_q/L , take one with exactly $T a_q$ nodes. The Gilboa–Peled every-integer-node-count theorem applied at degree $2m+2$ allows these choices, since $a_q \geq 1$. The standard interior Vandermonde regularization preserves moments through degree m and makes all nodes distinct and strictly interior: there are at least $m+2$ distinct support points in the higher-degree rule, hence at least m interior pivots, and the implicit-function argument removes endpoints and collisions. This argument is detailed in the reviewed one-exceptional proof and its every-size corollary.

Globally, the B rule has $KT = O((m+1)^6)$ equally likely nodes and the A rule has $LT = O((m+1)^{162})$ equally likely nodes. Their moments through degree m on every individual cluster are rational, being the integrals of rational polynomial densities over rational intervals. Every cluster contains at least m^2 distinct nodes. The new hybrid model is still fully fair: on a product of an A cluster and a B cluster, their relative order is fixed and every conditional old-order kernel is polynomial of degree at most m in each specified variable. The cellwise quadratures preserve its integral.

Shrink the cluster intervals slightly around these finitely many interior nodes to make their closures disjoint. The theorem in `clustered_multi_exceptional_lifting.tex` now applies with $k=2$. It retains the two displayed face counts and replaces the m uniforms by finite dice with one common face count. This proves the result. $\square$